

Rekindle: Enhancing Interactive Narrative in Virtual Reality with Coherent, Agency-driven Interactions

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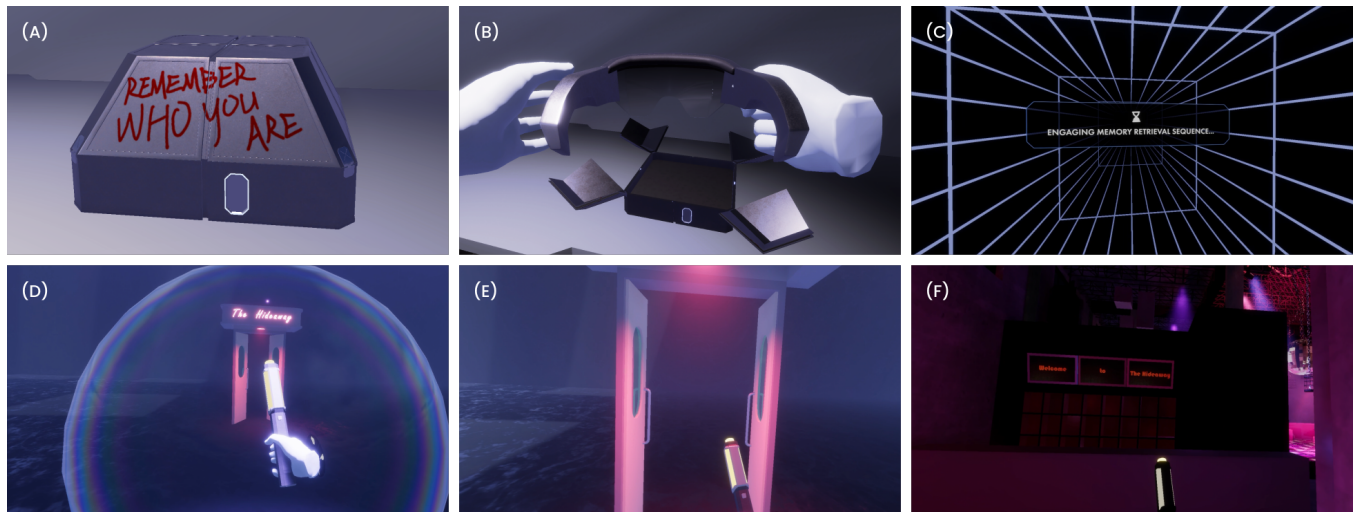


Figure 1: (A) A mysterious box containing the Mind Link device, (B) Mind Link, (C) Memory retrieval sequence, (D) Acquiring the Memory Illuminator in the Heedspace, (E) A portal to Memory Dimension, (F) Memory Dimension

ABSTRACT

In this research, we investigate the potential of enhanced interactivity in providing new possibilities for interactive narratives within Virtual Reality (VR). We introduce "Rekindle," a first-person narrative experience in VR. The narrative offers the interactor an embodied experience as a gay protagonist on a journey to reclaim lost memories, set against the backdrop of a dystopian future where a regime enforces stringent heteronormativity through the manipulation of memories. Central to this experience is the Memory Retrieval Mechanism, which empowers the interactor to explore and retrieve scattered memories. This activity reinforces immersion and the interactor's connection within the narrative. By integrating coherent interactions grounded in narrative, "Rekindle" facilitates the creation of dramatic agency, thereby enhancing the narrative by making the story more compelling and consistent.

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CCS CONCEPTS

• Applied computing → Media arts; • Human-centered computing → Virtual reality; Interaction paradigms.

KEYWORDS

Interactive Narrative, Virtual Reality, Dramatic Agency, Design, Game

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1 INTRODUCTION

Storytelling is so deeply entwined in human culture, with the power of storytelling, we can communicate complex information and allow people to see things from the world in a different perspective [4]. Storytelling has undergone transformative shifts with technological advancements. Traditional narratives, such as books, movies, and TV shows, while powerful, offer a linear and passive experience, as Janet Murray believed, this format might have reached its limit in conveying our world's complexities [7].

Interactive Digital Narratives (IDN), particularly within the realm of virtual reality, revolutionize this paradigm by immersing interactors directly into the narrative. This immersion not only grants interactors agency but also allows for interactive actions that are intrinsically linked to the narrative, thus fostering a profound sense of immersion and facilitating the "Active Creation of Belief."

However, many VR narratives have not fully harnessed the potential of this interactivity, employing traditional sequential narrative techniques. Our research seeks to more effectively leverage the interactivity in virtual reality. We aim to create dramatic agency by designing interactions that are both coherent and deeply grounded from narrative, thereby actualizing the concept of the "Active Creation of Belief." [9] Our research led to the creation of a first-person VR narrative experience "Rekindle" [7].

2 NARRATIVE CONTEXT

The narrative concept of "Rekindle" is motivated by fostering positive social change for the LGBTQ+ community. Although significant progress has been made in contemporary dialogues, inequality faced by the LGBTQ+ community persists globally. Our narrative is additionally informed by pivotal historical events, such as The Stonewall Riots and the practice of Conversion Therapy, which are critical to understanding queer history. VR narratives such as "Out of Exile" [5] and "Becoming Homeless: A Human Experience" [1] have demonstrated the potent capacity of interactive and immersive VR experiences to convey experiences and perspectives that might otherwise remain inaccessible to many. This capacity positions VR as an apt medium for engaging with and addressing issues related to the political suppression of the LGBTQ+ community.

In crafting the narrative of "Rekindle," we deliberately chose the science fiction genre, recognizing its dual capacity to both reflect societal issues, and to provide a creative latitude that facilitates the integration of narrative and interactivity. Science fiction, such as George Orwell's "1984," [11] effectively employs dystopian themes to underscore and critique contemporary societal concerns. Moreover, science fiction settings in various games have been adeptly integrated with interactions that become part of the narrative, strengthening the play's immersion. The "gravity gloves" in "Half-Life: Alyx" [16] function as a tool, allowing players to manipulate distant objects through a simulated "gravity force." This interaction, emblematic of science fiction technology, transcends conventional gameplay mechanics, serving not only to enhance playability but also to deepen the player's immersion in the game's dystopian future setting [6]. Similarly, "The Legend of Zelda: Tears of the Kingdom" [10] utilizes the "Purah Pad," a multi-purpose tool in the game that mirrors the shape of the game console Switch, thereby enhancing player immersion by blurring the lines between the game and reality. These examples underscore the effectiveness of integrating narrative and interactivity to enhance immersion and dramatic agency.

Informed by this, our narrative is strategically created to reflect the real-world suppression of the LGBTQ+ community, set in a dystopian future where a regime enforces stringent heteronormativity through the manipulation of memories. The interactor embodies a gay protagonist on a journey to recover lost memories. We incorporate the narrative setting of memory manipulation

by creating a "Memory Retrieval Mechanism." This is achieved by leveraging VR's interactivity and designing interactive artifacts that enhance the narrative with profound dramatic agency, weaving a compelling and coherent story. Among these artifacts is the "Mind Link," a tool enabling the player to transition between reality and the "Headspace," a virtual space where lost memories can be accessed. Additionally, the "Memory Illuminator" reveals erased memories scattered throughout the "Memory Dimension," which endowed the interactor with the capability to explore the space and relive the past. This mechanism not only sustained user engagement but also encouraged active exploration, enriching the narrative by creating a dramatic agency.

3 BACKGROUND STUDY

3.1 Traditional Narrative

In traditional storytelling mediums, narrators have crafted techniques to ensure audience adherence to predetermined narrative sequences, thereby maintaining the story structure and arc as envisioned by the storyteller. An example of such a technique is David Trottier's "The Magnificent 7 Plot Points," which elaborates on the classic Western three-act structure [15]. This methodological approach exemplifies the precision and deliberateness with which conventional storytelling mediums shape their narratives, guiding the audience through a carefully constructed sequence of events that align with the predetermined narrative trajectory.

3.2 Interactive Digital Narrative (IDN) in VR

Contrasting with traditional narratives, IDNs introduce pronounced interactivity and agency, elements usually absent in conventional storytelling. IDNs disrupt the linear narrative framework by offering a participatory experience, as highlighted by Murray. This interactive aspect enables interactors to navigate and potentially shape the narrative. VR narrative, an evolving medium, has adopted this participatory essence, as identified by Murray in 1997 [9]. Jeremy Bailenson contrasts traditional and VR narrative, suggesting that while traditional narratives emphasize narrator control, VR prioritizes interactor agency and exploration [2]. However, Chris Crawford noted that while traditional storytelling may lack the interactivity, it possesses a refined internal structure that IDNs might find challenging to replicate due to the inherent trade-offs with interactivity [4].

3.3 Research Challenge

In analyzing the current landscape of VR narratives, it becomes evident that numerous interactive VR experiences favor narrative over user agency. These narratives often incorporate traditional storytelling methods, such as montage and cutscenes, offering limited interactivity. However, the narrative progression tends to be minimally impacted by the user's engagement with these interactive components. For instance, the acclaimed "Book of Distance" [13] involves users in a character's role but constricts their narrative progression, despite the presence of interactive elements. As a result, interactor actions frequently have a negligible effect on the narrative's trajectory. This situation highlights a fundamental issue: the superficial nature of interactivity in these narratives, which fails

to enhance narrative depth, thus not fully realizing VR's potential for true interactive narrative.

Upon conducting an analytical examination of the relationship between narrative strength and user agency across a spectrum of storytelling mediums, a discernible pattern becomes evident:

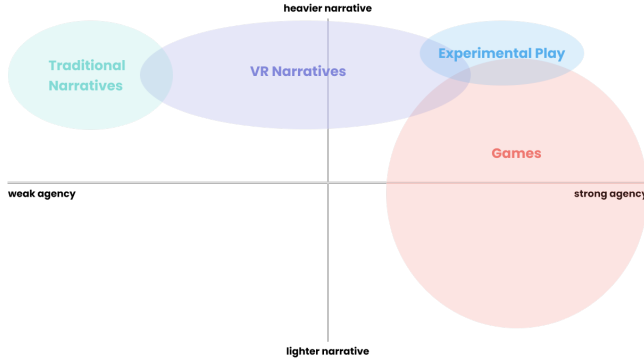


Figure 2: Narrative strength vs. agency quadrant

- Traditional narratives, such as films, TV series, and books, are situated in the upper-left quadrant, showcasing strong narratives with limited user agency.
- Most VR narratives are located towards the center-top, indicating strong narratives paired with moderate agency.
- Most games fall on the right side, representing a spectrum of agency from moderate to strong, with narratives varying from weak to strong.

This mapping suggests that, in most instances, there is a conflict between agency and narrative. However, there are outliers that defy these general classifications. For example, "Sleep No More," [12] an experimental immersive play, exemplifies a harmonious balance between agency and narrative. In this setup, the audience, metaphorically depicted as "ghosts," moves alongside actors across various settings, experiencing multiple narratives concurrently. This multi-sequential design empowers the audience with the agency to navigate over 100 different rooms and select storylines, resulting in a unique narrative journey for each participant.

3.4 Why VR

Embodiment and Interactivity: VR's inherent interactivity and immersion make it an exceptional medium for creating an embodied experience. Particularly in the context of fostering understanding towards the LGBTQ+ community, this immersive quality of VR makes it conducive to creating experiences where users can feel a direct, sensory connection to the narrative, thereby deepening their understanding.

Active Creation of Belief and Perspective-Taking: VR allows the interactor to intimately engage with the narrative and adopt the perspectives of characters, this "active creation of belief" and "perspective-taking" within VR encourages a profound connection to the character's experiences and stories, furthering the narrative's impact. [17]

The "gravity gloves" in "Half-Life: Alyx" serves as an exemplary illustration of maximizing VR's interactivity. In this game, the mechanic transcends traditional button-pressing actions; users physically perform a pulling gesture to interact with objects in the game world. The mapping of the pull's velocity to the gravity strength—where the force exerted by the interactor directly influences the distance objects move—demonstrates a sophisticated and intuitive use of VR interactivity to create a compelling and engaging interactive experience [6].

4 APPLICATION DESIGN

4.1 Design Goals

We argue that the objective extends beyond achieving a mere balance between narrative depth and user agency for an interactive narrative. The goal is to create an experience where narrative and agency reinforce each other. This presents a challenge: How might we more effectively leverage the interactivity afforded by IDN to create dramatic agency, thereby enriching the narrative?

In this project, our objective is to create a first-person narrative experience in VR where the interactor takes on the role of a gay protagonist. This focus encompasses two key aspects:

- (1) **Creating Dramatic Agency:** By leveraging interactivity within the narrative, the project aims to endow the user with meaningful agency, allowing for a significant impact on the story's progression and outcome.
- (2) **Fostering Active Creation of Belief:** We aim to design interactions that are not only logically and coherently intertwined with the narrative but also actively contribute to the interactor's belief in and engagement with the story. This ensures that interactions are not merely isolated events but are consistently integrated within the narrative, thereby enhancing the narrative's immersion.

4.2 Achieving Active Creation of Belief Through Coherent Interactions

The design ethos of this project rests on creating dramatic agency through coherent interactions that are grounded from narrative, naturally weaving together with the story [8], to achieve "Active Creation of Belief". To realize this, we introduced a three-layered virtual environment framework: "Reality", "Headspace" and "Memory Dimension," each playing a crucial role in creating a cohesive virtual experience, and the "Headspace" bridges reality with the memory dimension (Fig. 3).

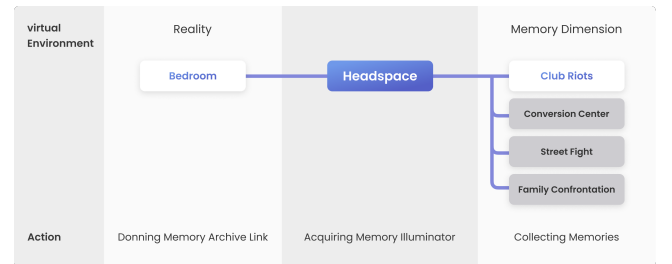


Figure 3: Three-layered virtual environment framework



Figure 4: (A)Memory particles, (B)Collecting memory, (C)Memory indicator, (D)Indicator's full

4.2.1 Reality.

The narrative journey begins in the "Reality," set within the protagonist's bedroom. Prompted by a mysterious box scrawled with "remember who you are" which contains the "Mind Link," a headset device for memory retrieval. Donning this headset initiates the memory retrieval sequence and transitions the interactor into the "Headspace." (Fig. 1: A, B, C)

4.2.2 Headspace.

"Headspace" symbolizes the protagonist's mental realm, a repository of lost memories. It functions as a nexus connecting the "Memory Dimension" corresponding to different storylines. Within this space, the interactor discovers the "Memory Illuminator," a key interactive element symbolizing the interactor's agency in their quest to reclaim lost memories. Acquiring the Memory Illuminator triggers a door to open (Fig. 1: D, E), leading the interactor into the "Memory Dimension," where they embark on a journey through distinct storylines to recover memories (Fig. 1: F).

4.2.3 Memory Dimension.

The "Memory Dimension" encompasses four unique virtual environments, each representing a different storyline: "Club Riots," "Conversion Center," "Street Fight," and "Family Confrontation." Each environment is accessible from the Headspace via individual portals. Within the Memory Dimension, interactors use the Memory Illuminator to uncover past events erased by the regime, visualized as animated particles that can be gathered by the Memory Illuminator.

In the present version of our prototype, the focus is narrowed to the "Club Riots" storyline. Within this story, the interactor is immersed in a vividly portrayed gay club scene, marked by its vibrant and seemingly free atmosphere. Here, the interactor engages in a romantic encounter with a character who later on provides the protagonist with the "Mind Link" device in the timeline after the protagonist's memory is altered by the government. However, this moment of personal revelation and connection is abruptly disrupted by chaotic riots within the club instigated by undercover law enforcement agents. Their objective is to detain individuals and subject them to forced memory manipulation.

This intricate design implementation ensures a deeply immersive and coherent experience, enabling interactors to actively engage with and influence the unfolding narrative, embodying the project's vision of Active Creation of Belief.

4.2.4 Transcendence.

Upon gathering all the memories, the portal to the Headspace will

open again, allowing the interactor to return to the headspace. The interactor's act of returning the Memory Illuminator triggers the memory particles to materialize as statues, symbolizing the protagonist's reclamation of memories. Upon gathering all the memories, the statues – emblematic of the memories themselves – become particles along with the entirety of the Headspace, drifting away. Following this transformation, the interactor is then transported back to the bedroom, signifying the interactor's metaphorical return to reality.

4.3 Creating Dramatic Agency with Interactivity

4.3.1 Memory Illuminator.

At the core of this VR experience is the Memory Retrieval Mechanism, embodied by the "Memory Illuminator" artifact, obtained within the Headspace. This interactive tool enables the interactor to identify, collect, and revisit gathered memories, endowing them with the capability to explore and uncover memories scattered throughout the virtual environment. Insights gathered from formative user tests conducted during the design process revealed a pronounced preference among interactors for a tangible iteration of the Memory Illuminator over a virtual menu projected on the interactor's hand, which aligns with the VR controller's functionality.

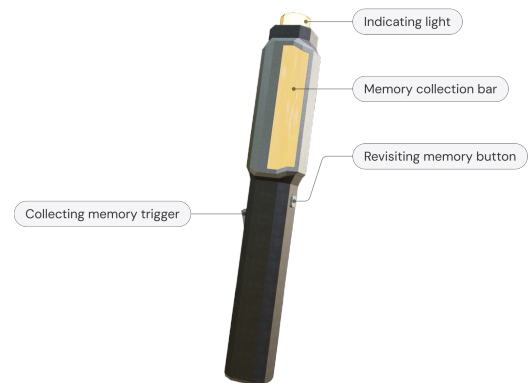


Figure 5: Memory Illuminator

The design of the Memory Illuminator's interface, specifically the "collecting memory trigger" and "revisiting memory button," is

meticulously aligned with the VR controller's button layout. This design choice fosters a sense of physical presence and interaction with the Memory Illuminator within the VR environment (Fig. 5). The tangible aspect of the Memory Illuminator's design draws inspiration from the Deluminator in the Harry Potter series, a magical device known for its light-extinguishing capabilities [14]. This tangible aspect not only renders the functionality more memorable but also significantly deepens the immersion experienced by the interactor, ultimately contributing to the creation of dramatic agency. This concept aligns with the design philosophy observed in the Universal Threshold Object, wherein tangible interactive elements are shown to enhance immersion and dramatic agency [3].

The mechanics serve two key purposes:

- Supporting user engagement
- Encouraging active exploration

4.3.2 Identifying Memory.

In the experience, approaching sites of past memories triggers their manifestation as animated particles, evoking the sensation of reliving these moments. The Memory Illuminator, when illuminated, serves as a visual cue, guiding interactors towards these memory site (Fig. 4: A).

4.3.3 Collecting Memory.

Upon activation of the "collecting memory trigger" by the interactor, memory particles are absorbed into the Memory Illuminator, with a bar indicator visually representing the number of memories collected. Once all memories are gathered, the bar indicator fully illuminates, signaling completion (Fig. 4: B, C, D).

4.3.4 Revisiting Memory.

Given that the memory pieces are scattered across various locations, each interactor's encounter with memories is distinct, resulting in a unique experiential trajectory. This may necessitate multiple revisitation of certain memories to fully comprehend the underlying events. To eliminate the need for physical re-navigation within the virtual environment, the Memory Illuminator allows for revisiting previously collected memories by pressing the "revisiting memory button." Each memory is represented as a hologram on the Illuminator, offering an intuitive and efficient method for interactors to engage with their collected memories.

5 CONCLUSION

This paper presents "Rekindle", an interactive narrative within VR, demonstrating how interactive models, when derived from and coherently integrated with the narrative, can facilitate clear interactions. This integration plays a pivotal role in empowering the interactor with dramatic agency, thereby enhancing the depth and richness of the narrative. Through the design of the interactive artifact "Memory Illuminator", the VR narrative empowers interactors to actively participate in the narrative through the meaningful action of collecting memories. This interactive model gives the interactor clear and instinctive interactions without verbal or textual instructions, creating an experience of dramatic agency, thereby enhancing the narrative depth. This work exemplifies VR's potential as a powerful storytelling medium, where narrative and user agency coexist and reinforce each other, creating an immersive and impactful experience.

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